SOME SOLUTIONS TO THE PLANETARY GEOSTROPHIC EQUATIONS

TWNH

Peterson and Callies (in review) (see also *Peterson and Callies* 2023) consider how bottom-intensified vertical mixing induces a basin-scale circulation in various simple non-flat-bottomed basins. They write equations for a bottom Ekman layer and couple them to an equation for the geostrophic contour vorticity balance, in the parlance of *Waldman and Giordani* (2023), which is a type of barotropic vorticity equation.

1. Governing Equations

The flow satisfies the (non-dimensional) planetary-geostrophic equations:

$$-fv = -\frac{\partial p}{\partial x} + \epsilon^2 \frac{\partial}{\partial z}\left(\nu \frac{\partial u}{\partial z}\right), \tag{1}$$

$$fu = -\frac{\partial p}{\partial y} + \epsilon^2 \frac{\partial}{\partial z}\left(\nu \frac{\partial v}{\partial z}\right), \tag{2}$$

(3)

(see Eqs. (9)–(11) of *Peterson and Callies* in review, Eq. (3.26) of *Klinger and Haine* 2019, and Eq. (2.274) of *Vallis* 2006). The boundary conditions are no flow at the bottom and a specified stress at the surface:

$$\text{Surface } z = 0 : \epsilon^2 \nu \partial_z (u, v) = (\tau^x, \tau^y), \tag{4}$$

$$\text{Bottom } z = -H : (u, v) = 0, \tag{5}$$

The flow is non-divergent:

$$\nabla \cdot \mathbf{u} = 0, \tag{6}$$

the buoyancy field is hydrostatic:

$$\frac{\partial p}{\partial z} = b, \tag{7}$$

and the buoyancy field satisfies the advection-diffusion-relaxation equation

$$\mu\rho\left(\frac{\partial b}{\partial t} + \mathbf{u}\cdot\nabla b\right) = \epsilon^2 \frac{\partial}{\partial z}\left(\kappa \frac{\partial b}{\partial z}\right) + \gamma\left(B - b\right). \tag{8}$$

(see Eqs. (11)–(13) of *Peterson and Callies* in review). The boundary conditions on buoyancy are:

$$\text{Surface } z = 0 : b = 0 \tag{9}$$

$$\text{Bottom } z = -H : \frac{\partial b}{\partial z} = 0, \tag{10}$$

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and the initial condition is $b = z$. The buoyancy field relaxes to a specified field $B(x, y, z, t)$ with rate γ.

The diffusivity and viscosity are given by

$$\kappa(z) = \kappa_0 + \exp\left(-\alpha(z + H)\right), \tag{11}$$

$$\nu = \text{constant} \tag{12}$$

Peterson and Callies (in review) Eq. (17).

2. Vertically-Integrated Flow

The vertically-integrated flow (U, V) is given by:

$$\begin{aligned} U &\equiv \int_{-H}^{0} u \, dz = -\int_{-H}^{0} z \partial_z u \, dz, \\ V &\equiv \int_{-H}^{0} v \, dz = -\int_{-H}^{0} z \partial_z v \, dz, \end{aligned} \tag{13}$$

exploiting the bottom boundary condition on (u, v). The equations for (U, V) come from vertically integrating the momentum equations (1), (2):

$$-fV = -\int_{-H}^{0} \frac{\partial p}{\partial x} \, dz + \epsilon^2 \int_{-H}^{0} \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \, dz \tag{14}$$

$$fU = -\int_{-H}^{0} \frac{\partial p}{\partial y} \, dz + \epsilon^2 \int_{-H}^{0} \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right) \, dz. \tag{15}$$

The depth-integrated pressure gradient in the x direction is:

$$\int_{-H}^{0} \frac{\partial p}{\partial x} \, dz = -\frac{\partial H}{\partial x} p(x, z = -H) + \frac{\partial}{\partial x} \int_{-H}^{0} p \, dz, \tag{16}$$

$$= -p_b \frac{\partial H}{\partial x} + \frac{\partial}{\partial x} \left[p_b H - \int_{-H}^{0} z \frac{\partial p}{\partial z} \right], \tag{17}$$

$$= H \frac{\partial p_b}{\partial x} - \frac{\partial}{\partial x} \int_{-H}^{0} z b \, dz, \tag{18}$$

using the Liebniz integration formula[1], where $p_b = p(x, -H(x))$ is the bottom pressure, and integration by parts[2]. The viscous term in the x direction is:

$$\int_{-H}^{0} \frac{\partial}{\partial z}\left(\nu \frac{\partial u}{\partial z}\right)\, dz = \left[\nu \frac{\partial u}{\partial z}\right]_{-H}^{0} = \frac{\tau^x}{\epsilon^2} - \nu \frac{\partial u}{\partial z}\bigg|_{z=-H}, \tag{19}$$

which uses the boundary condition (41).

Therefore (14) and (15) lead to the (steady, linear) barotropic momentum equations:

$$-fV = -H\frac{\partial p_b}{\partial x} - \frac{\partial \chi}{\partial x} - \tau_b^x + \tau^x, \tag{20}$$

$$fU = -H\frac{\partial p_b}{\partial y} - \frac{\partial \chi}{\partial y} - \tau_b^y + \tau^y, \tag{21}$$

where

$$\chi = -\int_{-H}^{0} zb\, dz \tag{22}$$

is the baroclinic potential energy, and $\epsilon^2\nu\, \partial_z(u,v)|_{z=-H} \equiv (\tau_b^x, \tau_b^y) = \tau_b$ is the bottom stress on the fluid (see *Waldman and Giordani* 2023 Eq. (9)).

An alternate form of these equations is

$$-fV = -H\frac{\partial p_s}{\partial x} - \int_{-H}^{0} \frac{\partial \lambda(z)}{\partial x}\, dz - \tau_b^x + \tau^x, \tag{23}$$

$$fU = -H\frac{\partial p_s}{\partial y} - \int_{-H}^{0} \frac{\partial \lambda(z)}{\partial y}\, dz - \tau_b^y + \tau^y, \tag{24}$$

$$\lambda(z) = \int_{z}^{0} b\, dz \tag{25}$$

is the vertically-integrated buoyancy and $p_s = p_b - \int_{-H}^{0} \lambda(z)\, dz$ is the surface pressure.

3. Barotropic Vorticity Equation, Barotropic Divergence Equation, and Geostrophic Contour Vorticity Balance

Waldman and Giordani (2023) show how to derive the barotropic vorticity equation and the geostrophic vorticity balance.

[1] The Liebniz integration formula reads:

$$\partial_x\left(\int_{a(x)}^{b(x)} f(x,y)\, dy\right) = \partial_x b f(x, b(x)) - \partial_x a f(x, a(x)) + \int_{a(x)}^{b(x)} \partial_x f(x,y)\, dy.$$

[2] Integration by parts gives:

$$\begin{aligned}\int_{-H}^{0} z\frac{\partial p}{\partial z}\, dz &= [zp]_{-H}^{0} - \int_{-H}^{0} p\, dz,\\ &= p_b H - \int_{-H}^{0} p\, dz.\end{aligned}$$

3.1. **Barotropic vorticity equation.** Take the curl of the barotropic momentum equations: $\partial_x(21)$-$\partial_y(20)$ to give:[3]

$$\partial_x \{fU\} - \partial_y \{-fV\} = \partial_x \left\{-H\frac{\partial p_b}{\partial y} - \frac{\partial \chi}{\partial y} - \tau_b^y + \tau^y\right\} - \partial_y \left\{-H\frac{\partial p_b}{\partial x} - \frac{\partial \chi}{\partial x} - \tau_b^x + \tau^x\right\}$$

$$\implies f\left(\partial_x U + \partial_y V\right) + \beta V = J(p_b, H) + \hat{\mathbf{z}} \cdot \left(\nabla \times (\tau - \tau_b)\right). \tag{27}$$

This is the balance of terms in the steady, linear barotropic vorticity equation (see *Waldman and Giordani* 2023 Eq. (10)). In this case, the horizontal divergence of the vertically-integrated flow vanishes. This constraint comes from (6), which implies that[4]

$$\int_{-H}^{0} \partial_x u + \partial_y v + \partial_z w \; dz = 0,$$
$$\implies \partial_x U + \partial_y V = 0. \tag{28}$$

Thus, for steady solutions we require that

$$J(p_b, H) + \hat{\mathbf{z}} \cdot \left(\nabla \times (\tau - \tau_b)\right) - \beta V = 0 \tag{29}$$

(see *Nøst and Isachsen* 2003 Eq. (13))

3.2. **Barotropic divergence equation.** Take the horizontal divergence of the barotropic momentum equations: $\partial_x(21)$-$\partial_y(20)$ to give:

$$\partial_x \{-fV\} + \partial_y \{fU\} = \partial_x \left\{-H\frac{\partial p_b}{\partial x} - \frac{\partial \chi}{\partial x} - \tau_b^x + \tau^x\right\} + \partial_y \left\{-H\frac{\partial p_b}{\partial y} - \frac{\partial \chi}{\partial y} - \tau_b^y + \tau^y\right\}$$

$$\implies f\left(\partial_x V - \partial_y U\right) - \beta U = \nabla p_b \cdot \nabla H + H\nabla^2 p_b + \nabla^2 \chi - \nabla \cdot (\tau - \tau_b). \tag{30}$$

This is the balance of terms in the steady, linear barotropic divergence equation, which also must be satisfied.

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[3]Recall that

$$\hat{\mathbf{z}} \cdot (\nabla \times \mathbf{U}) = \partial_x V - \partial_y U. \tag{26}$$

[4]Note that

$$\partial_x U = \partial_x \int_{H(x)}^{0} u(x,z)\; dz = \partial_x H\, u(x, -H(x)) + \int_{-H(x)}^{0} \partial_x u \; dz,$$

and

$$\int_{H(x)}^{0} \partial_z w(x,z)\; dz = w(x,0) - w(x,-H(x)).$$

The top boundary condition on the velocity is $w(x,0) = 0$ and the bottom boundary condition is $\mathbf{u} \cdot \nabla H = 0$, implying that

$$w(x, -H(x)) = -\left[u(x,-H(x))\partial_x H + v(x,-H(x))\partial_y H\right],$$

and therefore the vertically-integrated flow has zero horizontal divergence, which is (28).

3.3. **Geostrophic Contour Vorticity Balance.** Divide the barotropic momentum equations (20)–(21) by water depth H:

$$-\frac{fV}{H} = -\frac{\partial p_b}{\partial x} - \frac{1}{H}\frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H} \tag{31}$$

$$\frac{fU}{H} = -\frac{\partial p_b}{\partial y} - \frac{1}{H}\frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H}, \tag{32}$$

and take the curl

$$\begin{aligned}\frac{\partial}{\partial y}\left\{-\frac{fV}{H}\right\} - \frac{\partial}{\partial x}\left\{\frac{fU}{H}\right\} = \frac{\partial}{\partial y}\left\{-\frac{\partial p_b}{\partial x} - \frac{1}{H}\frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H}\right\}. \\ -\frac{\partial}{\partial x}\left\{-\frac{\partial p_b}{\partial y} - \frac{1}{H}\frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H}\right\},\end{aligned} \tag{33}$$

$$\begin{aligned}\implies \frac{\partial}{\partial y}\left\{-\frac{f}{H}\frac{\partial \Psi}{\partial x}\right\} + \frac{\partial}{\partial x}\left\{\frac{f}{H}\frac{\partial \Psi}{\partial y}\right\} = -\frac{\partial}{\partial y}\left\{\frac{1}{H}\frac{\partial \chi}{\partial x}\right\} + \frac{\partial}{\partial x}\left\{\frac{1}{H}\frac{\partial \chi}{\partial y}\right\} - \hat{\mathbf{z}}\cdot\left(\nabla \times \frac{\tau - \tau_b}{H}\right), \\ \implies -\frac{\partial}{\partial y}\left(\frac{f}{H}\right)\frac{\partial \Psi}{\partial x} + \frac{\partial}{\partial x}\left(\frac{f}{H}\right)\frac{\partial \Psi}{\partial y} = -\frac{\partial}{\partial y}\left(\frac{1}{H}\right)\frac{\partial \chi}{\partial x} + \frac{\partial}{\partial x}\left(\frac{1}{H}\right)\frac{\partial \chi}{\partial y} - \hat{\mathbf{z}}\cdot\left(\nabla \times \frac{\tau - \tau_b}{H}\right),\end{aligned} \tag{34}$$

$$\implies J\left(\frac{f}{H}, \Psi\right) = J\left(\frac{1}{H}, \chi\right) - \hat{\mathbf{z}}\cdot\left(\nabla \times \frac{\tau - \tau_b}{H}\right), \tag{35}$$

for streamfunction of the depth-integrated horizontal flow Ψ where

$$\begin{aligned}V &= \frac{\partial \Psi}{\partial x}, \\ U &= -\frac{\partial \Psi}{\partial y},\end{aligned} \tag{36}$$

and Jacobian $J(A, B) = \partial_x A\, \partial_y B - \partial_y A\, \partial_x B$. Notice how this recipe (vertically integrate, divide by H, then take the curl) eliminates the bottom pressure. The final result (35) is the same as Eq. (20) of *Peterson and Callies* (in review). It is called (a simplified form of) the **geostrophic contour vorticity balance** (GCVB) by *Waldman and Giordani* (2023), although *Peterson and Callies* (in review) call it the "barotropic vorticity equation". It is not the same equation as used by *Nøst and Isachsen* (2003) (which *Waldman and Giordani* (2023) call the "transport divergence vorticity balance").

Notice how the GCVB (35) is an equation for the streamfunction of the depth-integrated flow Ψ given the buoyancy field (and hence the JEBAR term) and the surface and bottom stresses. The GCVB shows how depth-integrated flow Ψ across f/H contours is driven by (i) the joint effect of baroclinicity and relief (JEBAR), (ii) curl of wind stress divided by H, (iii) curl of the bottom stress divided by H.

4. Solution to Frictional Geostrophic Equations in Steady State

Start from (1)–(2) and write the pressure as

$$p(x,y,z) = p_s(x,y) + \int_z^0 b(x,y,z')\,dz', \tag{37}$$

$$\implies \partial_x p(x,y,z) = \partial_x p_s(x,y) + \int_z^0 \partial_x b(x,y,z')\,dz' \tag{38}$$

$$\implies p_b(x,y) \equiv p(x,y,z=-H(x,y)) = p_s(x,y) + \int_{-H(x,y)}^0 b(x,y,z')\,dz', \tag{39}$$

where $p_s(x,y)$ is the surface pressure (and assuming that $p_s \ll p(-H)$). Solve these frictional geostrophic equations using a Green's function, defined so that $\mathfrak{u} = u + iv$, thus

$$\epsilon^2 \frac{\partial}{\partial z}\left(\nu \frac{\partial \mathfrak{u}}{\partial z}\right) - if\mathfrak{u} = \mathfrak{f}, \tag{40}$$

$$\text{Surface } z = 0 : \epsilon^2 \nu \partial_z \mathfrak{u} = \tau^x + i\tau^y, \tag{41}$$

$$\text{Bottom } z = -H : \mathfrak{u} = 0, \tag{42}$$

where $\mathfrak{f} = \frac{\partial p}{\partial x} + i\frac{\partial p}{\partial y}$ is the pressure gradient from (38).

The corresponding Green's function satisfies:

$$\epsilon^2 \frac{\partial}{\partial z}\left(\nu \frac{\partial G_{\mathfrak{u}}}{\partial z}\right) - ifG_{\mathfrak{u}} = \delta(z-\xi), \tag{43}$$

$$\text{Surface } z = 0 : \epsilon^2 \nu \partial_z G_{\mathfrak{u}} = 0, \tag{44}$$

$$\text{Bottom } z = -H : G_{\mathfrak{u}} = 0. \tag{45}$$

We find that the solution for constant viscosity $\nu = \nu_0$ is:

$$G_{\mathfrak{u}}(z;\xi) = \frac{i^{3/2}\mathrm{e}^{-\sqrt{i}\varphi(z+\xi)}}{2\nu_0\epsilon^2\varphi\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)} \begin{cases} -\mathrm{e}^{2\sqrt{i}\varphi\xi} + \mathrm{e}^{2\sqrt{i}\varphi(z+H(x))} + \mathrm{e}^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \le \xi, \\ -\mathrm{e}^{2\sqrt{i}\varphi z} + \mathrm{e}^{2\sqrt{i}\varphi(\xi+H(x))} + \mathrm{e}^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \ge \xi, \end{cases}, . \tag{46}$$

This expression implies the following results, which are used later:

$$G_{\mathfrak{u}}(z;0) = \frac{i^{3/2}\mathrm{e}^{-\sqrt{i}\varphi z}}{\nu_0\epsilon^2\varphi\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right))}\left[\mathrm{e}^{2\sqrt{i}\varphi(z+H(x))} - 1\right], \tag{47}$$

$$\int_{-H(x)}^0 G_{\mathfrak{u}}(z;\xi)\,d\xi = \frac{i\left(1+\mathrm{e}^{2\sqrt{i}\phi H(x)} - \mathrm{e}^{\sqrt{i}\phi(z+H(x))} - \mathrm{e}^{\sqrt{i}\phi(H(x)-z)}\right)}{\nu_0\epsilon^2\phi^2\left(1+\mathrm{e}^{2\sqrt{i}\phi H(x)}\right)} \tag{48}$$

$$\partial_z\, G_{\mathfrak{u}}(z;\xi)\big|_{z=-H(x)} = \frac{-\mathrm{e}^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0\epsilon^2\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}\left(1+\mathrm{e}^{2\sqrt{i}\varphi\xi}\right), \tag{49}$$

$$\partial_z\, G_{\mathfrak{u}}(z;0)\big|_{z=-H(x)} = \frac{-2\mathrm{e}^{\sqrt{i}\varphi H(x)}}{\nu_0\epsilon^2\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}, \tag{50}$$

where $\varphi = \sqrt{(f/\nu_0)}/\epsilon$ (see `Example_barotropic_GeostrophicFlow_v0.2.ipynb` and `ExampleSolution_v0.5.ipynb` #1.1, 1.2, and note that this parameter contains f so is a function of space in principle). And then

$$\mathfrak{u}(z;x) = \int_{-H(x)}^{0} G_{\mathfrak{u}}(z;\xi)\left[\partial_x p(x,\xi) + i\partial_y p(x,\xi)\right]\, d\xi - G_{\mathfrak{u}}(z;0)\left(\tau^x + i\tau^y\right). \tag{51}$$

The baroclinic pressure (due to b) adds to the barotropic pressure (due to $p_s(x,y)$) in the integrand of this expression. Otherwise, everything else is known (remember that x in these expressions stands for (x,y) in general). Work on the expression (51) for $\mathfrak{u}$ using (38) some more, to avoid the multiple vertical integrals:

$$\begin{aligned}
\mathfrak{u}(z;x) &= \int_{-H(x)}^{0} G_{\mathfrak{u}}(z;\xi)\left[(\partial_x + i\partial_y)\left(p_s(x) + \int_{\xi}^{0} b(x,z')\, dz'\right)\right]\, d\xi - G_{\mathfrak{u}}(z;0)\left(\tau^x + i\tau^y\right),\\
&= (\partial_x + i\partial_y)\, p_s(x)\int_{-H(x)}^{0} G_{\mathfrak{u}}(z;\xi)\, d\xi\\
&\quad + \int_{-H(x)}^{0} G_{\mathfrak{u}}(z;\xi)\left[(\partial_x + i\partial_y)\int_{\xi}^{0} b(x,z')\, dz'\right]\, d\xi\\
&\quad - G_{\mathfrak{u}}(z;0)\left(\tau^x + i\tau^y\right).
\end{aligned}\tag{52}$$

The final term is given by (47) and the integral on the second line is given by (48). The integral on the third line is dealt with in section 6.1 for a specific relaxation buoyancy field.

Also, the bottom current gradient is given by

$$\begin{aligned}
-\partial_z\, \mathfrak{u}(z;x)\big|_{z=-H(x)} &= \int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i}\varphi(\xi - H(x))}}{\nu_0\epsilon^2\left(1 + \mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}\left(1 + \mathrm{e}^{2\sqrt{i}\varphi\xi}\right)\left[\partial_x p(x,\xi) + i\partial_y p(x,\xi)\right]\, d\xi\\
&\quad - \frac{2\mathrm{e}^{\sqrt{i}\varphi H(x)}}{\nu_0\epsilon^2\left(1 + \mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}\left(\tau^x + i\tau^y\right),
\end{aligned}\tag{53}$$

using (49) and (50), and therefore the bottom stress τ_b on the fluid is

$$\begin{aligned}
\tau_b = -\nu_0\epsilon^2\partial_z\, \mathfrak{u}(z;x)\big|_{z=-H(x)} &= \int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i}\varphi(\xi - H(x))}}{1 + \mathrm{e}^{2\sqrt{i}\varphi H(x)}}\left(1 + \mathrm{e}^{2\sqrt{i}\varphi\xi}\right)\left[\partial_x p(x,\xi) + i\partial_y p(x,\xi)\right]\, d\xi\\
&\quad - \frac{2\mathrm{e}^{\sqrt{i}\varphi H(x)}}{1 + \mathrm{e}^{2\sqrt{i}\varphi H(x)}}\left(\tau^x + i\tau^y\right).
\end{aligned}\tag{54}$$

Notice how the derivatives of the surface pressure part of p end up multiplying expressions that are functions of x.

Work on the bottom stress expression some more by substituting for the pressure gradient from (38):

$$\begin{aligned}
\tau_b = &\int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i}\varphi(\xi - H(x))}}{1 + \mathrm{e}^{2\sqrt{i}\varphi H(x)}}\left(1 + \mathrm{e}^{2\sqrt{i}\varphi\xi}\right)\left[(\partial_x + i\partial_y)\left(p_s(x) + \int_{\xi}^{0} b(x,z')\, dz'\right)\right]\, d\xi\\
&- \frac{2\mathrm{e}^{\sqrt{i}\varphi H(x)}}{1 + \mathrm{e}^{2\sqrt{i}\varphi H(x)}}\left(\tau^x + i\tau^y\right).
\end{aligned}\tag{55}$$

The integral of the surface pressure term can be computed:

$$\int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i\varphi}(\xi-H(x))}}{1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}}\left(1+\mathrm{e}^{2\sqrt{i\varphi}\xi}\right)\left[\left(\partial_x+i\partial_y\right)p_s(x)\right]\,d\xi = \left[\left(\partial_x+i\partial_y\right)p_s(x)\right]\int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i\varphi}(\xi-H(x))}}{1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}}\left(1+\mathrm{e}^{2\sqrt{i\varphi}\xi}\right)\,d\xi$$

$$= \frac{1}{\sqrt{i\varphi}}\tanh\left(\sqrt{i\varphi}H(x)\right)\left[\left(\partial_x+i\partial_y\right)p_s(x)\right]. \tag{56}$$

Thus,

$$\tau_b = \frac{1}{\sqrt{i\varphi}}\tanh\left(\sqrt{i\varphi}H(x)\right)\left[\left(\partial_x+i\partial_y\right)p_s(x)\right] + \int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i\varphi}(\xi-H(x))}}{1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}}\left(1+\mathrm{e}^{2\sqrt{i\varphi}\xi}\right)\left[\left(\partial_x+i\partial_y\right)\int_{\xi}^{0} b(x,z')\,dz'\right]\,d\xi$$

$$- \frac{2\mathrm{e}^{\sqrt{i\varphi}H(x)}}{1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}}\left(\tau^x+i\tau^y\right),$$

$$\tau_b = S(x)\left[\left(\partial_x+i\partial_y\right)p_s(x)\right] + T(x), \tag{57}$$

where

$$S(x) = \frac{1}{\sqrt{i\varphi}}\tanh\left(\sqrt{i\varphi}H(x)\right),$$

$$T(x) = \int_{-H(x)}^{0} \frac{\mathrm{e}^{-\sqrt{i\varphi}(\xi-H(x))}}{1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}}\left(1+\mathrm{e}^{2\sqrt{i\varphi}\xi}\right)\left[\left(\partial_x+i\partial_y\right)\int_{\xi}^{0} b(x,z')\,dz'\right]\,d\xi - \frac{2\mathrm{e}^{\sqrt{i\varphi}H(x)}}{1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}}\left(\tau^x+i\tau^y\right). \tag{58}$$

For an unstratified, flat-bottomed fluid ($\chi = 0, \nabla p_b = \nabla p_s, H = H_0$), specifying the surface pressure gradient ∇p_s closes the problem. We compute the surface stress from (55) by substituting (54). Then $u(z), v(z), U, V$ and the force balance can all be computed. See Figure 1 for an example calculation.. Alternatively, the surface stress can be specified, and the pressure gradient can be computed (see `Example_barotropic_GeostrophicFlow_v0.3.ipynb`). In practice, it seems that the bottom stress always vanishes. Why?? Is this correct??

5. Frictional Geostrophic Equations Closure with the Barotropic Momentum Equation

The problem is closed using the barotropic momentum equation, as follows (*Salmon* 1998, p. 170). This determines the unknown surface pressure field given the surface stress (and vice versa). In particular, work from (20)–(21),

$$\mathfrak{U}(z;x) \equiv U + iV = \int_{-H(x)}^{0} \mathfrak{u}(z;x)\,dz, \tag{59}$$

$$= \int_{-H(x)}^{0}\left\{\int_{-H(x)}^{0} G_{\mathfrak{u}}(z;\xi)\left[\partial_x p(x,\xi) + i\partial_y p(x,\xi)\right]\,d\xi - G_{\mathfrak{u}}(z;0)\left(\tau^x+i\tau^y\right)\right\}\,dz. \tag{60}$$

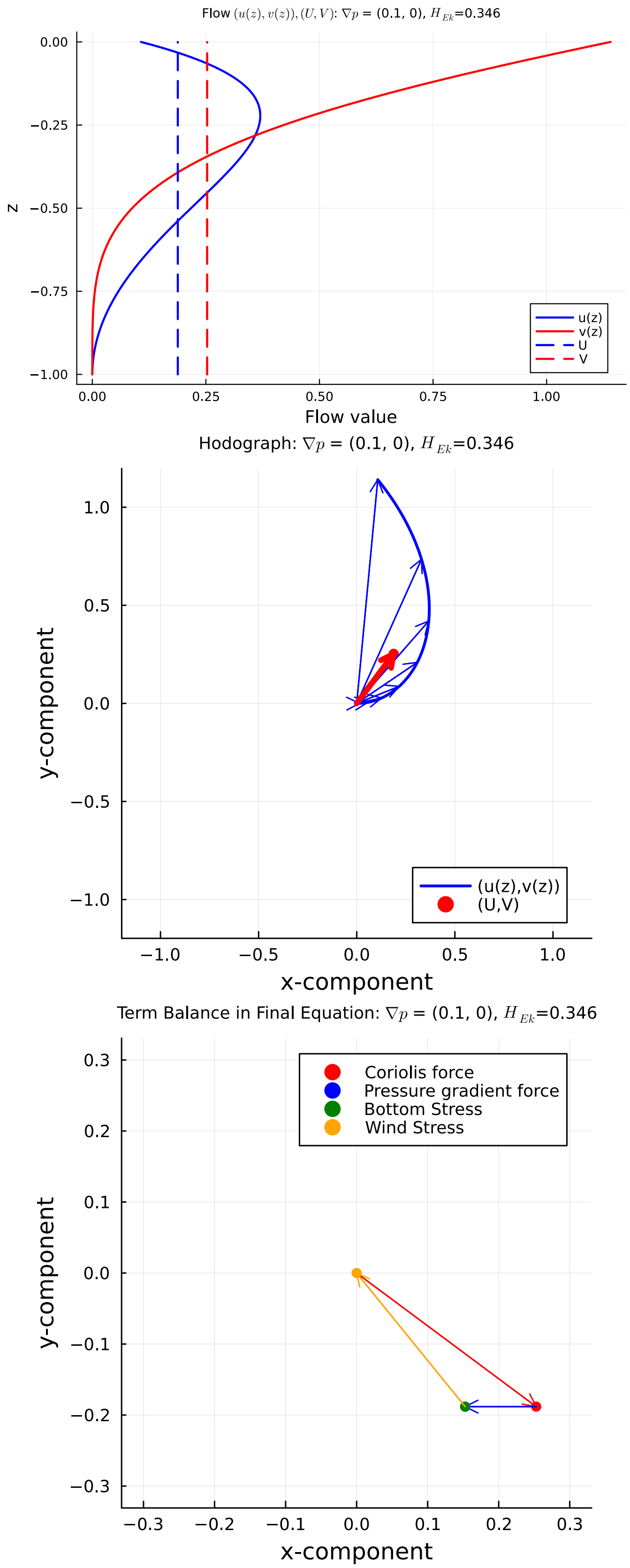

FIGURE 1. Example solution to the barotropic frictional geostrophic flow problem from `Example_barotropic_GeostrophicFlow_v0.2.ipynb`. See also Figure 2.12 from *Vallis* (2006).

Computing the z integral of the Green's function $G_{\mathtt{u}}(z;\xi)^5$ gives

$$\int_{-H(x)}^{0} G_{\mathtt{u}}(z;\xi)\;dz = \frac{-i\mathrm{e}^{-\sqrt{i}\varphi\xi}\left(\mathrm{e}^{\sqrt{i}\varphi\xi}-\mathrm{e}^{\sqrt{i}\varphi H(x)}\right)\left(\mathrm{e}^{\sqrt{i}\varphi(\xi+H(x))}-1\right)}{\nu_0\epsilon^2\varphi^2\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)},$$

$$\Longrightarrow \int_{-H(x)}^{0} G_{\mathtt{u}}(z;0)\;dz = \frac{-i\left(\mathrm{e}^{\sqrt{i}\varphi H(x)}-1\right)^2}{\nu_0\epsilon^2\varphi^2\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}. \tag{61}$$

Therefore,

$$\mathfrak{U}(x) = \frac{-i}{f\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}\times$$
$$\left\{\int_{-H(x)}^{0} \mathrm{e}^{-\sqrt{i}\varphi\xi}\left(\mathrm{e}^{\sqrt{i}\varphi\xi}-\mathrm{e}^{\sqrt{i}\varphi H(x)}\right)\left(\mathrm{e}^{\sqrt{i}\varphi(\xi+H(x))}-1\right)\left[\partial_x p(x,\xi)+i\partial_y p(x,\xi)\right]\;d\xi\right.$$
$$\left. + \left(\mathrm{e}^{\sqrt{i}\varphi H(x)}-1\right)^2(\tau^x+i\tau^y)\right\} \tag{62}$$

(see `PlanetaryGeostrophicTheory_v0.2.ipynb`). Substituting for the pressure gradient from (37) then gives:

$$\mathfrak{U}(x) = \frac{-i}{f\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}\times$$
$$\left\{\int_{-H(x)}^{0} \mathrm{e}^{-\sqrt{i}\varphi\xi}\left(\mathrm{e}^{\sqrt{i}\varphi\xi}-\mathrm{e}^{\sqrt{i}\varphi H(x)}\right)\left(\mathrm{e}^{\sqrt{i}\varphi(\xi+H(x))}-1\right)\left[(\partial_x+i\partial_y)\left(p_s(x)+\int_{\xi}^{0} b(x,z')\;dz'\right)\right]\;d\xi\right.$$
$$\left. + \left(\mathrm{e}^{\sqrt{i}\varphi H(x)}-1\right)^2(\tau^x+i\tau^y)\right\}. \tag{63}$$

The integral of the surface pressure term can be computed:

$$\int_{-H(x)}^{0} \mathrm{e}^{-\sqrt{i}\varphi\xi}\left(\mathrm{e}^{\sqrt{i}\varphi\xi}-\mathrm{e}^{\sqrt{i}\varphi H(x)}\right)\left(\mathrm{e}^{\sqrt{i}\varphi(\xi+H(x))}-1\right)\left[(\partial_x+i\partial_y)\,p_s(x)\right]\;d\xi$$
$$= \left[(\partial_x+i\partial_y)\,p_s(x)\right]\left[-H(x)\left(\mathrm{e}^{2\sqrt{i}\varphi H(x)}+1\right)+\frac{\mathrm{e}^{2\sqrt{i}\varphi H(x)}-1}{\sqrt{i}\varphi}\right]$$

[5]This presumes (correctly) that the denominator in (46) does not vanish.

(see `ExampleSolution_v0.2.ipynb`). Therefore,

$$\begin{aligned}
\mathfrak{U}(x) = &\frac{-i}{f\left(1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}\right)}\times \\
&\left\{\left[(\partial_x+i\partial_y)\,p_s(x)\right]\left[-H(x)\left(\mathrm{e}^{2\sqrt{i\varphi}H(x)}+1\right)+\frac{\mathrm{e}^{2\sqrt{i\varphi}H(x)}-1}{\sqrt{i\varphi}}\right]+\right. \\
&\int_{-H(x)}^{0}\mathrm{e}^{-\sqrt{i\varphi}\xi}\left(\mathrm{e}^{\sqrt{i\varphi}\xi}-\mathrm{e}^{\sqrt{i\varphi}H(x)}\right)\left(\mathrm{e}^{\sqrt{i\varphi}(\xi+H(x))}-1\right)\left[(\partial_x+i\partial_y)\int_{\xi}^{0}b(x,z')\,dz'\right]\,d\xi \\
&\left.+\left(\mathrm{e}^{\sqrt{i\varphi}H(x)}-1\right)^2(\tau^x+i\tau^y)\right\}
\end{aligned}$$

$$\implies \mathfrak{U}=U+iV=A(x)\;(\partial_x+i\partial_y)\,p_s(x)+B(x), \tag{64}$$

where

$$\begin{aligned}
A(x) &= \frac{-i}{f\left(1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}\right)}\left[-H(x)\left(\mathrm{e}^{2\sqrt{i\varphi}H(x)}+1\right)+\frac{\mathrm{e}^{2\sqrt{i\varphi}H(x)}-1}{\sqrt{i\varphi}}\right], \\
A(x) &= \frac{i}{f}\left[H(x)+\frac{1-\mathrm{e}^{2\sqrt{i\varphi}H(x)}}{\sqrt{i\varphi}\left(1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}\right)}\right], \\
B(x) &= \frac{-i}{f\left(1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}\right)}\left[\int_{-H(x)}^{0}\mathrm{e}^{-\sqrt{i\varphi}\xi}\left(\mathrm{e}^{\sqrt{i\varphi}\xi}-\mathrm{e}^{\sqrt{i\varphi}H(x)}\right)\left(\mathrm{e}^{\sqrt{i\varphi}(\xi+H(x))}-1\right)\left[(\partial_x+i\partial_y)\int_{\xi}^{0}b(x,z')\,dz'\right]\,d\xi\right. \\
&\quad \left.+\left(\mathrm{e}^{\sqrt{i\varphi}H(x)}-1\right)^2(\tau^x+i\tau^y)\,,\right]
\end{aligned} \tag{65}$$

(see `ExampleSolution_v0.2.ipynb` to verify these expressions, and notice that this $B(x)$ is not the same as the relaxation buoyancy profile $B(z)$ used in section 6).

This formula expresses the depth-integrated flow in terms of the applied surface stress and the buoyancy field (in $B(x)$) and the surface pressure field. The $A(x)$ field only depends on the fluid depth $H(x)$. Note also that the streamfunction for the depth-integrated flow (36) is given by:

$$\begin{aligned}
\nabla^2\Psi &= -\left(i\partial_x+\partial_y\right)\mathfrak{U}, \\
&= -\left(i\partial_x+\partial_y\right)\left[A(x)\;(\partial_x+i\partial_y)\,p_s(x)+B(x)\right].
\end{aligned} \tag{66}$$

To close the problem, utilize the constraint that the vertically-integrated flow is non-divergent (6). This requirement amounts to (from (29)):

$$\Re\left\{(\partial_x-i\partial_y)\,\mathfrak{U}(x)\right\}=0,$$

which implies that

$$\Re\big\{\,(\partial_x-i\partial_y)\left[A(x)\;(\partial_x+i\partial_y)\,p_s(x)+B(x)\right]\,\big\}=0$$

from (64). The final equation for the surface pressure is thus

$$
\begin{aligned}
&\Re\big\{\partial_x\left[A(x)\partial_x p_s(x)+iA(x)\partial_y p_s(x)\right]-i\partial_y\left[A(x)\partial_x p_s(x)+iA(x)\partial_y p_s(x)\right]\big\}=-\Re\big\{\left(\partial_x-i\partial_y\right)B(x)\big\}\\
&\qquad\implies \Re\big\{\partial_x A(x)\partial_x p_s(x)+A(x)\partial_{xx}p_s(x)+i\partial_x A(x)\partial_y p_s(x)-\\
&\qquad\qquad i\partial_y A(x)\partial_x p_s(x)+\partial_y A(x)\partial_y p_s(x)+A(x)\partial_{yy}p_s(x)\big\}=-\Re\big\{\left(\partial_x-i\partial_y\right)B(x)\big\}.
\end{aligned}
\tag{67}
$$

This equation is a 2nd-order elliptic equation for $p_s(x)$ given the surface stress and the buoyancy field. In more compact form, this is the real parts of:

$$
A\left(\partial_{xx}+\partial_{yy}\right)p_s+\partial_x A\partial_x p_s+\partial_y A\partial_y p_s+i\left(\partial_x A\partial_y p_s-\partial_y A\partial_x p_s\right)=-\left(\partial_x-i\partial_y\right)B \tag{68}
$$

To write the weak form, which is used to compute p_s solutions numerically, take the divergence form of the equation, multiply by test function $v(x)$, and integrate over all space:

$$
\begin{aligned}
&\int_{x,y} v\left(\partial_x-i\partial_y\right)\left[A\ \left(\partial_x+i\partial_y\right)p_s\right]\ dy\ dx=\int_{x,y} v\left(-\partial_x+i\partial_y\right)B\ dy\ dx,\\
\implies &\int_{x,y}\left(\partial_x-i\partial_y\right)v\left[A\ \left(\partial_x+i\partial_y\right)p_s\right]\ dy\ dx=\int_{x,y} B\left(-\partial_x+i\partial_y\right)v\ dy\ dx
\end{aligned}
\tag{69}
$$

assuming that p_s and v vanish on the boundaries. Similarly, the weak form to find the streamfunction for the depth-integrated flow, Ψ, is (from (66)):

$$
\begin{aligned}
&\int_{x,y} v\nabla^2\Psi\ dy\ dx=\int_{x,y} v\left(-\left(i\partial_x+\partial_y\right)\left[A(x)\ \left(\partial_x+i\partial_y\right)p_s(x)+B(x)\right]\right)\ dy\ dx,\\
\implies &\int_{x,y}\nabla\Psi\cdot\nabla v\ dy\ dx=-\int_{\Omega} C(x)\left(i\partial_x+\partial_y\right)v\ dy\ dx,
\end{aligned}
$$

where

$$
C(x)=A(x)(\partial_x+i\partial_y)p_s(x)+B(x).
$$

assuming that Ψ and v vanish on the boundaries.

6. Solution to Linear Buoyancy Equation in Steady State

At least for steady state and negligible buoyancy advection, the buoyancy equation (8) reads:

$$
\epsilon^2\frac{\partial}{\partial z}\left(\kappa\frac{\partial b}{\partial z}\right)+\gamma\left(B(z)-b\right)=0, \tag{70}
$$

$$
\text{Surface } z=0: b=0 \tag{71}
$$

$$
\text{Bottom } z=-H: \frac{\partial b}{\partial z}=0 \tag{72}
$$

for buoyancy profile $B(z)$ (in principle, B could depend on (x,y) too) and relaxation coefficient γ, and vertical buoyancy gradient at the sea bed is zero. The internal buoyancy source allows for steady-state solutions that have non-trivial solutions (the $t\to\infty$ solution to (8)–(10) is vanishing b everywhere; the *Peterson and Callies* in review solutions are transient, relaxing to zero). This problem (70) can be solved exactly using a Green's function, at least for constant diffusivity, and I suspect also for diffusivity given by (11).

In detail: Solve the Green's function problem first:

$$\frac{\epsilon^2}{\gamma}\frac{\partial}{\partial z}\left(\kappa\frac{\partial G_b}{\partial z}\right) - G_b = -\delta(z-\xi), \tag{73}$$

$$\text{Surface } z=0: G_b = 0, \tag{74}$$

$$\text{Bottom } z=-H: \frac{\partial G_b}{\partial z} = 0. \tag{75}$$

This is a second order linear ordinary differential equation with two boundary conditions, which can be solved exactly. The solution for constant diffusivity $\kappa = \kappa_0$ is:

$$G_b(z;\xi) = \frac{\mathrm{e}^{\psi(z+\xi)}}{2\kappa_0\psi\left(\mathrm{e}^{2\psi H(x)}+1\right)}\begin{cases}\left(1-\mathrm{e}^{-2\psi\xi}\right)\left(\mathrm{e}^{2\psi H(x)}+\mathrm{e}^{-2\psi z}\right) & \text{for } z\le\xi,\\ \left(1-\mathrm{e}^{-2\psi z}\right)\left(\mathrm{e}^{2\psi H(x)}+\mathrm{e}^{-2\psi\xi}\right) & \text{for } z\ge\xi,\end{cases} \tag{76}$$

with $\psi = \sqrt{\gamma/\kappa_0}/\epsilon$. Therefore, given $G_b(z;\xi)$,

$$b(z;x) = -\gamma\int_{-H(x)}^{0} G_b(z;\xi)B(\xi;x)\,d\xi. \tag{77}$$

See `DiffusionEquationSolution.ipynb` and `FindBuoyancyField_and_GeostrophicFlow_v0.?.ipynb` and `ExampleSolution_v0.5.ipynb`.

In solving the momentum equations, one needs horizontal derivatives of the vertically-integrated buoyancy field to compute the pressure gradient (38). We need this term:

$$\begin{aligned}\int_z^0 \partial_x b(x,y,z')\,dz' &= -\gamma\int_z^0\partial_x\int_{-H(x)}^0 G_b(z';\xi)B(\xi;x)\,d\xi\,dz',\\ &= -\gamma\partial_x\int_{-H(x)}^0\left[\int_z^0 G_b(z';\xi)\,dz'\right]B(\xi;x)\,d\xi.\end{aligned}$$

The integral in square brackets can be computed analytically. One finds that:

$$\begin{aligned}&\int_z^0 G_b(z';\xi)\,dz' = \frac{1}{2\kappa_0\psi^2\epsilon^2\left(\mathrm{e}^{2\psi H(x)}+1\right)}\Big[\\ &\mathrm{e}^{\psi(z-\xi+2H(x))} - \mathrm{e}^{\psi(z+\xi+2H(x))} + 2\mathrm{e}^{\psi(\xi+2H(x))} + 2\mathrm{e}^{-\xi\psi} + \mathrm{e}^{\psi(\xi-z)} - \mathrm{e}^{-\psi(z+\xi)} -\\ &\mathrm{e}^{\psi(\xi+2H(x)-\mathrm{Max}(z,\xi))} - \mathrm{e}^{-\psi(\xi-2H(x)-\mathrm{Max}(z,\xi))} - \mathrm{e}^{\psi(\xi-\mathrm{Max}(z,\xi))} - \mathrm{e}^{-\psi(\xi-\mathrm{Max}(z,\xi))}\Big]\end{aligned}$$

(see `ExampleSolution_v0.5.ipynb`, #1.1).

6.1. **Example linear $B(z)$ profile.** For a linear $B(z)$ function, $B(z) = \alpha z$, we have (from (77)):

$$\begin{aligned}\frac{b(x,y,z')}{-\alpha\gamma} &= \int_{-H(x)}^0 G_b(z;\xi)\xi\,d\xi\\ &= \frac{1}{\kappa_0\psi^2\epsilon^2}\left[\frac{\mathrm{e}^{\psi H(x)}\left(\mathrm{e}^{\psi z}-\mathrm{e}^{-\psi z}\right)}{\psi\left(\mathrm{e}^{2\psi H(x)}+1\right)} - z\right]\end{aligned} \tag{78}$$

(see `ExampleSolution_v0.5.ipynb` #2.7). Also,

$$\int_z^0 \partial_x b(x,y,z')\, dz' = \frac{-\alpha\gamma}{2\kappa_0\psi^2\epsilon^2}\,\partial_x \int_{-H(x)}^0 \frac{\xi}{\mathrm{e}^{2\psi H(x)}+1}\Big[\mathrm{e}^{\psi(z-\xi+2H(x))} - \mathrm{e}^{\psi(z+\xi+2H(x))} + 2\mathrm{e}^{\psi(\xi+2H(x))} + 2\mathrm{e}^{-\xi\psi} + \mathrm{e}^{\psi(\xi-z)} - \mathrm{e}^{-\psi(z+\xi)} - \mathrm{e}^{\psi(\xi+2H(x)-\mathrm{Max}(z,\xi))} - \mathrm{e}^{-\psi(\xi-2H(x)-\mathrm{Max}(z,\xi))} - \mathrm{e}^{\psi(\xi-\mathrm{Max}(z,\xi))} - \mathrm{e}^{-\psi(\xi-\mathrm{Max}(z,\xi))}\Big]\, d\xi.$$

Now,

$$\int_{-H(x)}^0 \frac{\xi}{\mathrm{e}^{2\psi H(x)}+1}\Big[\mathrm{e}^{\psi(z-\xi+2H(x))} - \mathrm{e}^{\psi(z+\xi+2H(x))} + 2\mathrm{e}^{\psi(\xi+2H(x))} + 2\mathrm{e}^{-\xi\psi} + \mathrm{e}^{\psi(\xi-z)} - \mathrm{e}^{-\psi(z+\xi)} - \mathrm{e}^{\psi(\xi+2H(x)-\mathrm{Max}(z,\xi))} - \mathrm{e}^{-\psi(\xi-2H(x)-\mathrm{Max}(z,\xi))} - \mathrm{e}^{\psi(\xi-\mathrm{Max}(z,\xi))} - \mathrm{e}^{-\psi(\xi-\mathrm{Max}(z,\xi))}\Big]\, d\xi = z^2 + \frac{4\mathrm{e}^{\psi H(x)} - 2\mathrm{e}^{\psi(z+H(x))} - 2\mathrm{e}^{\psi(H(x)-z)}}{\psi^2\left(\mathrm{e}^{2\psi H(x)}+1\right)}$$

(see `ExampleSolution_v0.5.ipynb` #1.2). Hence,

$$\begin{aligned}
\int_z^0 \partial_x b(x,y,z')\, dz' &= \frac{-\alpha\gamma}{2\kappa_0\psi^2\epsilon^2}\,\partial_x\left[z^2 + \frac{4\mathrm{e}^{\psi H(x)} - 2\mathrm{e}^{\psi(z+H(x))} - 2\mathrm{e}^{\psi(H(x)-z)}}{\psi^2\left(\mathrm{e}^{2\psi H(x)}+1\right)}\right],\\
&= \frac{-\alpha\gamma}{2\kappa_0\psi^2\epsilon^2}\partial_x\left[\frac{4\mathrm{e}^{\psi H(x)} - 2\mathrm{e}^{\psi(z+H(x))} - 2\mathrm{e}^{\psi(H(x)-z)}}{\psi^2\left(\mathrm{e}^{2\psi H(x)}+1\right)}\right].\\
&= \frac{\alpha\gamma}{\kappa_0\psi^3\epsilon^2}\,\partial_x H(x)\frac{\left(\mathrm{e}^{\psi H(x)} - \mathrm{e}^{3\psi H(x)}\right)\left(\mathrm{e}^{z\psi} - 2 + \mathrm{e}^{-z\psi}\right)}{\left(\mathrm{e}^{2\psi H(x)}+1\right)^2} \qquad (79)
\end{aligned}$$

(see `ExampleSolution_v0.5.ipynb` #1.3). This expression is needed in the computation of the horizontal velocity from (51). We need to compute the part of the velocity driven by the baroclinic pressure gradient, namely, the baroclinic part of:

$$\int_{-H(x)}^0 G_{\mathtt{u}}(z;\xi)\,\partial_x p(x,\xi)\, d\xi, \qquad (80)$$

and similarly for the y component of the pressure gradient. From (38),

$$\partial_x p(x,y,z) = \partial_x p_s(x,y) + \int_z^0 \partial_x b(x,y,z')\, dz'$$

Thus, the baroclinic part of (80) is

$$\int_{-H(x)}^0 G_{\mathtt{u}}(z;\xi) \int_\xi^0 \partial_x b(x,y,z')\, dz'\, d\xi.$$

(see also (52)). Hence, we need to compute

$$\frac{\alpha\gamma}{\kappa_0\psi^3\epsilon^2}\frac{\mathrm{e}^{\psi H(x)} - \mathrm{e}^{3\psi H(x)}}{\left(\mathrm{e}^{2\psi H(x)}+1\right)^2}\,\partial_x H(x)\int_{-H(x)}^0 G_{\mathtt{u}}(z;\xi)\left(\mathrm{e}^{\xi\psi} - 2 + \mathrm{e}^{-\xi\psi}\right)\, d\xi$$

from (79). The key calculation is

$$
\begin{aligned}
&\int_{-H(x)}^{0} G_{\mathtt{u}}(z;\xi)\mathrm{e}^{\xi\psi}\, d\xi = \\
&\frac{\left(-i^{\frac{3}{2}}\psi\mathrm{e}^{\psi H(x)} + i^{\frac{3}{2}}\psi\mathrm{e}^{2\sqrt{i}z\phi+\psi H(x)+2\sqrt{i}\phi H(x)} - \phi\mathrm{e}^{\sqrt{i}\phi(2z+H(x))} - \phi\mathrm{e}^{\sqrt{i}\phi H(x)} + \phi\mathrm{e}^{z\psi+\sqrt{i}z\phi+\psi H(x)} + \phi\mathrm{e}^{z\psi+\sqrt{i}z\phi+\psi H(x)+2\sqrt{i}\phi H(x)}\right)}{\nu_0\phi\epsilon^2\left(\psi^2\mathrm{e}^{2\sqrt{i}\phi H(x)} + \psi^2 - i\phi^2\mathrm{e}^{2\sqrt{i}\phi H(x)} - i\phi^2\right)} \\
&= \frac{\left[i^{\frac{3}{2}}\psi\mathrm{e}^{\psi H(x)}\left(\mathrm{e}^{2\sqrt{i}\phi(z+H(x))}-1\right) - \phi\mathrm{e}^{\sqrt{i}\phi H(x)}\left(\mathrm{e}^{2\sqrt{i}\phi z}+1\right) + \phi\mathrm{e}^{z\psi+\sqrt{i}z\phi+\psi H(x)}\left(1+\mathrm{e}^{2\sqrt{i}\phi H(x)}\right)\right]\mathrm{e}^{-\sqrt{i}z\phi-\psi H(x)}}{\nu_0\phi\epsilon^2\left(\psi^2-i\phi^2\right)\left(\mathrm{e}^{2\sqrt{i}\phi H(x)}+1\right)}
\end{aligned}
$$

$$
= \frac{1}{\nu_0\epsilon^2\left(\psi^2-i\phi^2\right)}\left\{\frac{\left[i^{\frac{3}{2}}\psi\mathrm{e}^{\psi H(x)}\left(\mathrm{e}^{2\sqrt{i}\phi(z+H(x))}-1\right) - \phi\mathrm{e}^{\sqrt{i}\phi H(x)}\left(\mathrm{e}^{2\sqrt{i}\phi z}+1\right)\right]\mathrm{e}^{-\sqrt{i}z\phi-\psi H(x)}}{\phi\left(\mathrm{e}^{2\sqrt{i}\phi H(x)}+1\right)} + \mathrm{e}^{z\psi}\right\}. \tag{81}
$$

(see `ExampleSolution_v0.5.ipynb` #0.3 and the test comparison for `the_pbarog_term` using numerical quadrature).

Finally, the two z integrals can be performed to compute the $T(x)$ and $B(x)$ functions from (58) and (65), respectively. For $T(x)$ we need (in addition to the stress term)

$$
\begin{aligned}
&\int_{-H(x)}^{0}\frac{\mathrm{e}^{-\sqrt{i}\varphi(\xi-H(x))}}{1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}}\left(1+\mathrm{e}^{2\sqrt{i}\varphi\xi}\right)\left[\left(\partial_x+i\partial_y\right)\int_{\xi}^{0} b(x,z')\, dz'\right]\, d\xi \\
&= \frac{\alpha\gamma\left(\mathrm{e}^{\left(\psi+\sqrt{i}\varphi\right)H(x)} - \mathrm{e}^{\left(3\psi+\sqrt{i}\varphi\right)H(x)}\right)}{\kappa_0\psi^3\epsilon^2\left(1+\mathrm{e}^{2\psi H(x)}\right)^2\left(1+\mathrm{e}^{2\sqrt{i}\varphi H(x)}\right)}\left(\partial_x+i\partial_y\right)H(x)\times \\
&\left[\frac{1-\mathrm{e}^{-\left(\psi-\sqrt{i}\varphi\right)H(x)}}{\psi-\sqrt{i}\varphi} + \frac{1-\mathrm{e}^{-\left(\psi+\sqrt{i}\varphi\right)H(x)}}{\psi+\sqrt{i}\varphi} + \frac{1-\mathrm{e}^{\left(\psi+\sqrt{i}\varphi\right)H(x)}}{-\psi-\sqrt{i}\varphi} + \frac{1-\mathrm{e}^{\left(\psi-\sqrt{i}\varphi\right)H(x)}}{-\psi+\sqrt{i}\varphi}\right. \\
&\left.+\frac{2}{\sqrt{i}\varphi}\left(\mathrm{e}^{-\sqrt{i}\varphi H(x)} - \mathrm{e}^{\sqrt{i}\varphi H(x)}\right)\right]
\end{aligned}
$$

(see `ExampleSolution_v0.5.ipynb` #1.4).

For $B(x)$ we need this term:

$$\int_{-H(x)}^{0} \mathrm{e}^{-\sqrt{i\varphi}\xi}\left(\mathrm{e}^{\sqrt{i\varphi}\xi}-\mathrm{e}^{\sqrt{i\varphi}H(x)}\right)\left(\mathrm{e}^{\sqrt{i\varphi}(\xi+H(x))}-1\right)\left[(\partial_x+i\partial_y)\int_{\xi}^{0} b(x,z')\,dz'\right]\,d\xi$$
$$=\frac{\alpha\gamma\left(\mathrm{e}^{\psi H(x)}-\mathrm{e}^{3\psi H(x)}\right)}{\kappa_0\psi^3\epsilon^2\left(1+\mathrm{e}^{2\psi H(x)}\right)^2}\left(\partial_x+i\partial_y\right)H(x,y)\times$$
$$\left\{\mathrm{e}^{\sqrt{i\varphi}H(x)}\left[\frac{1-\mathrm{e}^{-\left(\psi+\sqrt{i\varphi}\right)H(x)}}{\psi+\sqrt{i\varphi}}+\frac{1-\mathrm{e}^{-\left(\psi-\sqrt{i\varphi}\right)H(x)}}{\psi-\sqrt{i\varphi}}+\frac{1-\mathrm{e}^{-\left(\sqrt{i\varphi}-\psi\right)H(x)}}{\sqrt{i\varphi}-\psi}+\frac{1-\mathrm{e}^{\left(\psi+\sqrt{i\varphi}\right)H(x)}}{-\psi-\sqrt{i\varphi}}\right.\right.$$
$$\left.+\frac{\mathrm{e}^{\sqrt{i\varphi}H(x)}}{\psi}\left(\mathrm{e}^{-\psi H(x)}-\mathrm{e}^{\psi H(x)}\right)+\frac{2}{\sqrt{i\varphi}}\left(\mathrm{e}^{-\sqrt{i\varphi}H(x)}-\mathrm{e}^{\sqrt{i\varphi}H(x)}\right)\right]$$
$$\left.+\frac{\mathrm{e}^{-\psi H(x)}}{\psi}-\frac{\mathrm{e}^{\psi H(x)}}{\psi}+2\left(1+\mathrm{e}^{2\sqrt{i\varphi}H(x)}\right)H(x,y)\right\} \tag{82}$$

(see `ExampleSolution_v0.5.ipynb` #1.5).

7. Example Solutions: Work in Progress

Next steps are to solve for:

(1) $b=0, H(x,y)=1-x^2-y^2$ in a circular parabolic basin. See `ExampleSolution_v0.3.ipynb`. DONE
(2) $b(x,y)$ given by solving the vertical diffusive buoyancy equation (see section 6) and $H(x,y)=1-x^2-y^2$ in a circular parabolic basin. DONE. For `SymPy` version, see `ExampleSolution_0.5.ipynb`, for `Symbolics` version, see `ExampleSolution_0.6.ipynb`. Note that the talk for the November 2025 Arctic Dynamics Workshop in Paris used `ExampleSolution_0.4.ipynb`, which has an error in the computation of $\mathfrak{u}(z)$.
(3) Add computation of Ψ. See `ExampleSolution_0.7.ipynb`. DONE.
(4) Work example with non-trivial bathymetry, like a basin separated by ridge. See `ExampleSolution_0.8.ipynb` DONE.
(5) Add variable f. Work mid-latitude β-plane example with flat bottom.
(6) Re-dimensionalize.

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